

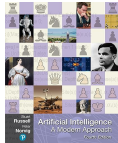


MSML610: Advanced Machine Learning

Lesson 01.2: The Foundations of AI

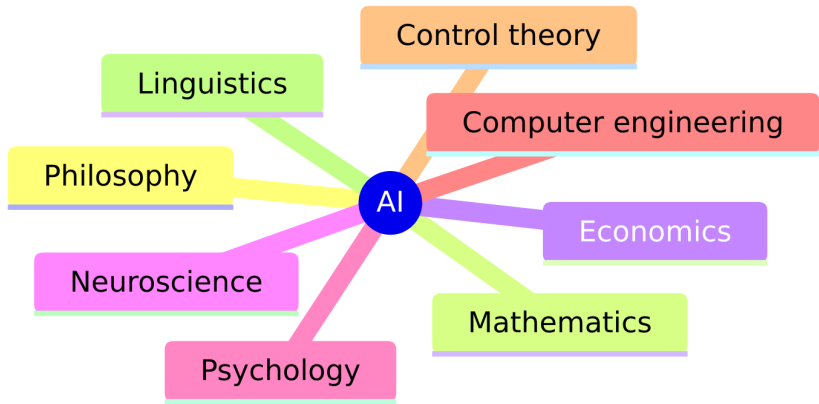
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References: - AIMA (Artificial Intelligence: a Modern Approach), Chap 1



- *The Foundations of AI*

AI Relates to Many Other Disciplines



AI and Philosophy (1/2)

- **Can formal rules be used to draw valid conclusions?**
 - **Reasoning**
 - * Logic studies rules of proper reasoning
 - * Aristotle (400 BCE) formulated laws governing the rational mind
 - * Machines were built for arithmetic operations (e.g., Pascaline, 1600)
 - **Rationalism**
 - * Use reasoning to understand the world
- **How does the mind arise from a physical brain?**
 - **Dualism**
 - * Nature follows physical laws
 - * Part of the human mind (“the soul”) is exempt from physical laws
 - **Materialism**
 - * The mind is a physical system, following the laws of physics
 - * Where is free will? Free will is the perception of available choices

AI and Philosophy (2/2)

- **What does knowledge come from?**

- **Empiricism**

- * Knowledge acquired via senses
 - * E.g., learn that trees are green by looking at them

- **Induction**

- * General rules from associations
 - * E.g., many swans are white, infer all swans are white

- **Logical Positivism**

- * Knowledge as logical theories linked to observations
 - * E.g., scientific hypotheses connected to experimental data



- **How does knowledge lead to action?**

- **Utilitarianism**

- * Actions justified by logic connecting goals and outcomes

- **Consequentialism**

- * Right or wrong determined by action's expected outcomes

AI and Cognitive Psychology

- **How do humans think and act?**
 - **Cognitive psychology**
 - * Brain is an information-processing device
 - * Stimuli translated into internal representation
 - * Representation manipulated by cognitive processes to derive new internal representations (“beliefs”)
 - * Representations turned into actions (“goals”)
 - **Cognitive science**
 - * Use computer models to address memory, language, and logic thinking
 - * Dual / opposite of AI
 - **Human-computer interaction**
 - * Computers augment human abilities
 - * From artificial intelligence (AI) to intelligence augmentation (IA)

AI and Mathematics

- **What are the formal rules to draw valid conclusions?**
 - **Formal logic**
 - * Logical deduction rules (Boole, 1850)
 - * First-order logic includes objects and relations (Frege, 1879)
 - **Limits to deduction**
 - * Some statements are “undecidable”
 - * Incompleteness theorem: in any formal theory true statements exist that cannot be proved (Godel, 1931)
- **How do we reason with uncertain information?**
 - **Probability**
 - * Mathematics of uncertainty
 - * Cardano, Pascal, Bernoulli, Bayes (1500-1700)
 - **Statistics**
 - * Combines data with probability
 - * E.g., experiment design, data analysis, hypothesis testing, asymptotics

AI and Economics (1/2)

- **How to make decisions to maximize payoff given preferences?**
 - **Economies**
 - * Agents maximize economic well-being (utility)
 - * Studies desires and preferences
 - **Decision theory**
 - * Making decisions under uncertainty for preferred outcomes
 - * Probability theory + utility theory
 - * E.g., investment choices, policy decisions
- **How to make decisions when payoffs are result of several actions?**
 - **Operations research**
 - * Make rational decisions with payoffs for sequence of actions (Bellman, 1957)
 - * E.g., Markov Decision Processes
 - **Satisficing**
 - * Decisions that are good enough
 - * Closer to human behavior
 - * E.g., choosing a restaurant that meets basic criteria rather than finding the perfect one

AI and Economics (2/2)

- **How multiple agents with different goals act?**
 - **Large economies**
 - * Many agents with no mutual impact
 - * Ignore other agents' actions
 - * E.g., national economy where individual actions don't affect market
 - **Small economies**
 - * One player's actions influence others' utility
 - * E.g., local market where one seller's pricing affects competitors
 - **Game theory**
 - * Small economies resemble a “game” (Von Neumann, 1944)
 - * Rational agents might need randomized strategies
 - * E.g., rock-paper-scissors where randomization prevents predictability

- **How can you create systems that understand natural language?**
 - **Computational linguistics** (NLP)
 - * Studies sentence structure and meaning
 - * Machine translation (e.g., Google Translate)
 - * Sentiment analysis in social media
 - * Automated customer support chatbots
- **How does language relate to thought?**
 - **Knowledge representation**
 - * How to represent knowledge for computer reasoning
 - * E.g., first order knowledge, knowledge graphs

AI and Neuroscience

■ Brain

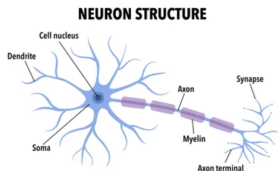
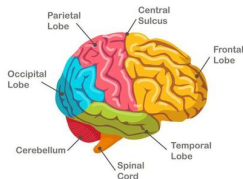
- Parts handle specific cognitive functions
- Information processing in the cerebral cortex
- E.g., frontal lobe injury may impair decision-making

■ Anatomy of the brain

- Composed of ~100 billion neurons
 - * Each neuron connects with 10-100k others via synapses
 - * Axons enable long-range connections
- Signals propagate through electrochemical reactions
- Short-term pathways support long-term connections (learning)

■ Memory

- No theory yet about individual memory storage
- Current theory: memories reconstructed



The Brain Causes the Mind

- **Simple cells lead to thought and consciousness**
 - **Truly amazing!**
 - Complex processes emerge
- Supercomputers' complexity rivals the brain
- **Brain-machine interface**
 - Brain adjusts to devices
 - E.g., learn to use prosthetics as limbs
- **AI singularity**
 - Future point when AI surpasses human intelligence
 - * AI improves autonomously, leading to rapid growth
 - * Recursive self-improvement leads to superintelligence
 - Potential societal impact
 - * Control problem/value alignment: ensure AI aligns with human values
 - * Economic/social disruption due to automation
 - Achieving brain's intelligence level remains unknown

AI and Computer Science

- **What can be computed?**
 - **Algorithm**
 - * Procedure to solve problems
 - * E.g., algorithm for computing GCD (Euclid, 300 BCE)
 - **Limits to computation**
 - * Turing machine (1936): computes any computable function
 - * Some functions are non-computable
 - * E.g., the halting problem, i.e., decide if a program terminates
 - **Tractability**
 - * Complexity classes: polynomial vs exponential complexity
 - * Problem is intractable if solving time grows exponentially with size
 - * P vs NP

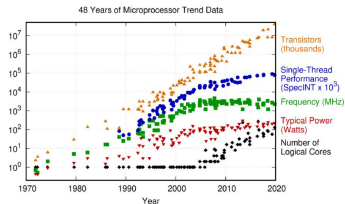
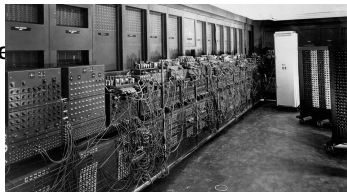
AI and Control Theory

- **How can artifacts operate under their own control?**
 - **Control theory**
 - * Study self-regulating feedback control systems
 - * E.g., a water regulator that maintains a constant water flow
 - * Mechanisms to minimize error between current and goal states
 - * Kalman Filter (Kalman, 1960)
 - * Based on calculus, matrix, stochastic optimal control
 - * AI: logical inference, symbolic planning, computation

AI and Computer Engineering

How can we build an efficient computer

- **Electronic computers**
 - * Built during World War II
- **Moore's Law**
 - * Performance doubled every 18 months (1970-2005)
 - * Power and scaling issues shifted focus to multi-core over clock speed
- **Hardware for AI**
 - * GPUs
 - * TPUs
 - * Wafer-scale engines
- **Current Trends**
 - * Massive parallelism (like brain function)
 - * Computing power doubling every 3 months
 - * GPUs / TPUs used in deep



CPU



GPU



TPU